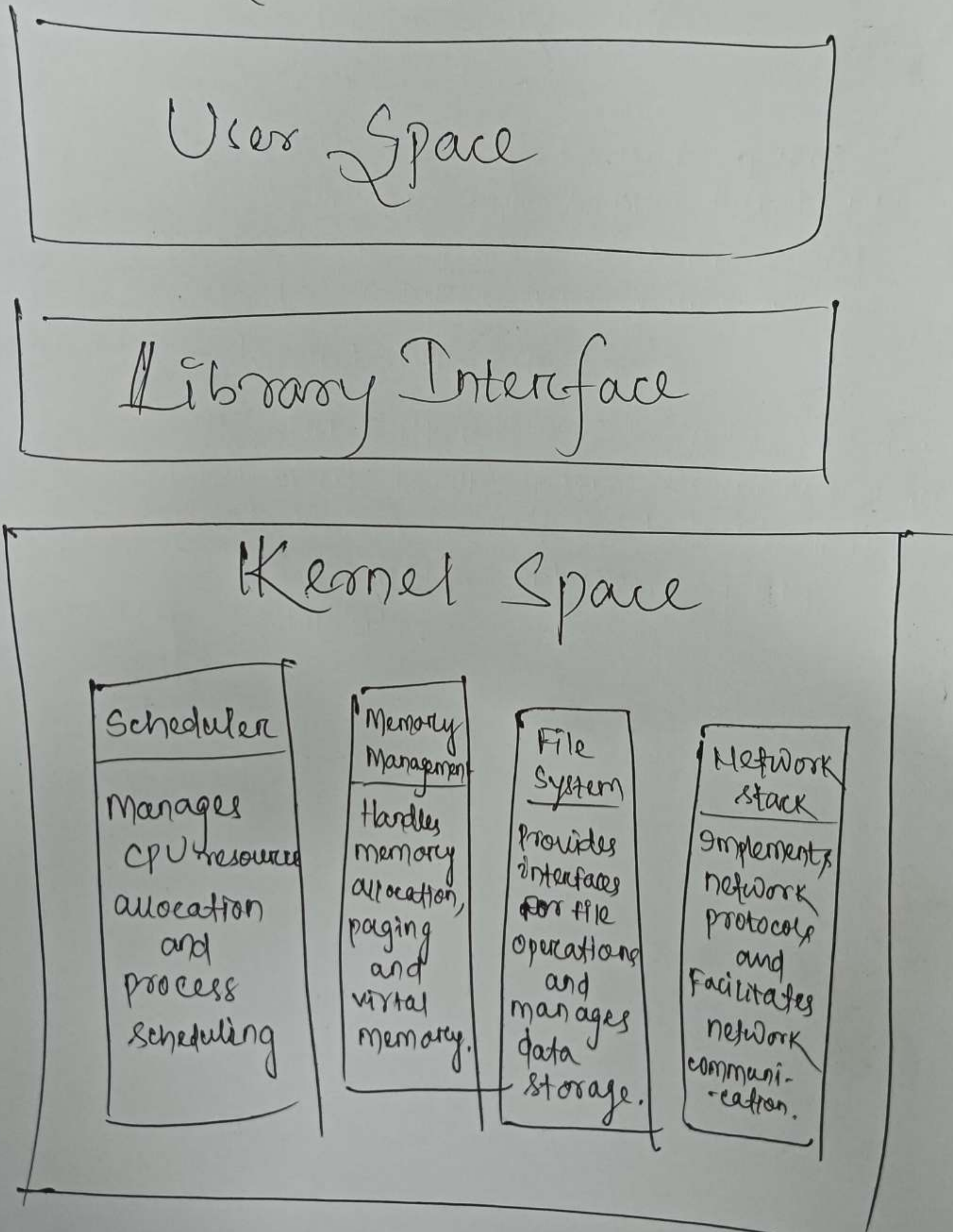
Linux Kernel Architecture



Short Descriptions of Linux Kernel Components :-

1) Process Schedular :-

The schedular manages process execution by determining which process runs at every given time. It ensures fairness and efficiency, supporting various scheduling algorithms for multitasking and time-sharing.

2) Memory Management Unit (MMU) :-

This component handles the allocation and deallocation of memory to processes. It supports virtual memory, page swapping, and memory protection to ensure stability and performance.

3) File System :-

The file system manages how data is stored, organized and retrieved on storage devices. It supports various formats like ext4, FAT and NTFS, and provides abstraction for file operations via system calls.

4) Network Stack :-

The network stack enables communication over networks using standard protocols like TCP/IP. It handles packet routing, error checking and ensures reliable data transmission between devices.

5) Inter-Process Communication (IPC)

IPC mechanisms like pipes, shared memory, message queues, and semaphores allow processes to communicate and synchronize their actions. This is crucial for multitasking and complex application execution.

6) Device Drivers :-

Device drivers acts as translators between the hardware and the Kernel. Each driver provides a standard interface for the OS to interact with hardware devices such as printers, disks or USBs.

7) System call Interface (SCI)

The SCI provides a bridge for user-space applications to interact with Kernel functions. It translates user requests into Kernel-level instructions for resource access and control.

8) Kernel mode and user mode :-

The Linux Kernel operates in Kernel mode with unrestricted access to all system resources, whereas user applications run in user mode with limited access. This separation enhances system security & stability.